
BUSINESS INTELLIGENCE AND DATA SUMMARISATION USING AI PROMPTS

Domain: *Business Intelligence / AI*

Data Understanding

- **Business Domain – Finance / Economics**
- The dataset that contains numerical data related to key economic and financial indicators.
- It likely includes variables such as economic growth measures, market performance indicators, interest rates, inflation levels, and other macroeconomic factors.
- The data appears to be structured in tabular format (rows and columns), making it suitable for financial analysis, trend analysis, forecasting, and business intelligence reporting.
- The dataset contains:
 - **Total Rows:** 3000
 - **Total Columns:** 24
 - Each row represents financial and economic data recorded for a specific date.
 - Each column represents a different financial or macroeconomic indicator.

Economic Indicators	Values
• Date	The specific date of the recorded data
• Stock Index	Overall market index value
• Open Price	Opening price of the stock/index
• Close Price	Closing price of the stock/index
• Daily High	Highest price of the day
• Daily Low	Lowest price of the day
• Trading Volume	Total shares traded
• GDP Growth (%)	Economic growth rate
• Inflation Rate (%)	Inflation percentage
• Unemployment Rate (%)	Unemployment percentage

• Interest Rate (%)	Central bank interest rate
• Consumer Confidence Index	Consumer sentiment indicator
• Government Debt (Billion USD)	Total government debt
• Corporate Profits (Billion USD)	Corporate earnings
• Forex USD/EUR –	Exchange rate (USD to EUR)
• Forex USD/JPY	Exchange rate (USD to JPY)
• Crude Oil Price (USD per Barrel)	Oil market price
• Gold Price (USD per Ounce)	Gold market price
• Real Estate Index	Real estate market performance
• Retail Sales (Billion USD)	Retail market sales
• Bankruptcy Rate (%)	Business bankruptcy percentage
• Mergers & Acquisitions Deals	Number of M&A deals
• Venture Capital Funding (Billion USD)	VC investment amount
• Consumer Spending (Billion USD)	Total consumer expenditure



Data Cleaning & Preparation



Before Data Cleaning

1. Missing (null) values were present in some columns.
2. Duplicate records existed in the dataset.
3. Date column was not properly formatted.
4. Some numerical columns were stored as text.
5. Inconsistent formatting and extra spaces were found.
6. Outliers were present in financial values like stock prices and trading volume.

Steps Followed in Data Cleaning

1. Identified missing values using filters and sorting.

◆ **Tool Used:**

- **Filter Option → Filter blanks**
- **Go To Special → Blanks**

2. Handled missing values by filling (mean/median) or removing rows where necessary.

◆ **Method 1: Replace with Average**

For numeric columns like GDP, Inflation:

=AVERAGE(B:B)

Then use:

- **Fill missing cells with the calculated average.**

◆ **Method 2: Remove Rows**

- **Select dataset**
- **Go to Data → Filter → Filter Blanks → Delete Rows**

3. Removed duplicate records using the “Remove Duplicates” option.

◆ **Tool Used:**

- **Data → Remove Duplicates**

4. Converted the Date column into proper date format.

If date is in text format:

=DATEVALUE(A2)

Then format as Date:

- **Right Click → Format Cells → Date**

5. Changed text-formatted numbers into numeric format.

◆ **Method 1:**

- **Select column → Click warning sign → Convert to Number**

◆ **Method 2 (Formula):**

=VALUE(A2)

6. Standardized percentage and currency formats.

Select column → Home → % (Percentage Style) OR

Right Click → Format Cells → Percentage

7. Checked and verified outliers in financial columns.

8. Removed extra spaces and ensured consistent formatting.

=TRIM(A2)

After Data Cleaning

1. Dataset became free from duplicates.
2. Missing values were handled properly.
3. All columns had correct data types.
4. Date format was standardized.
5. Data became accurate, consistent, and ready for analysis.
6. KPI calculations and visualizations became more reliable.



Business KPI

- **GDP Growth:** Shows whether the economy is growing or slowing down.
- **Inflation Rate:** Shows how fast prices are increasing.
- **Commodity Prices:** Show changes in oil and gold prices that affect business costs.
- **Consumer Spending:** Shows how much people are spending in the market.



KPI List

➤ Market KPIs

- Stock Index Performance
- Daily Market Return
- Market Volatility
- Trading Volume Trend

➤ Economic KPIs

- GDP Growth Rate
- Inflation Rate
- Unemployment Rate
- Interest Rate Trend
- Consumer Confidence Index

➤ Financial KPIs

- Corporate Profit Growth
- Government Debt Level
- Bankruptcy Rate
- Mergers & Acquisitions Activity
- Venture Capital Funding Trend

➤ Commodity & Global KPIs

- Crude Oil Price Trend
- Gold Price Trend
- Exchange Rate Movement (USD/EUR, USD/JPY)
- Real Estate Index Growth

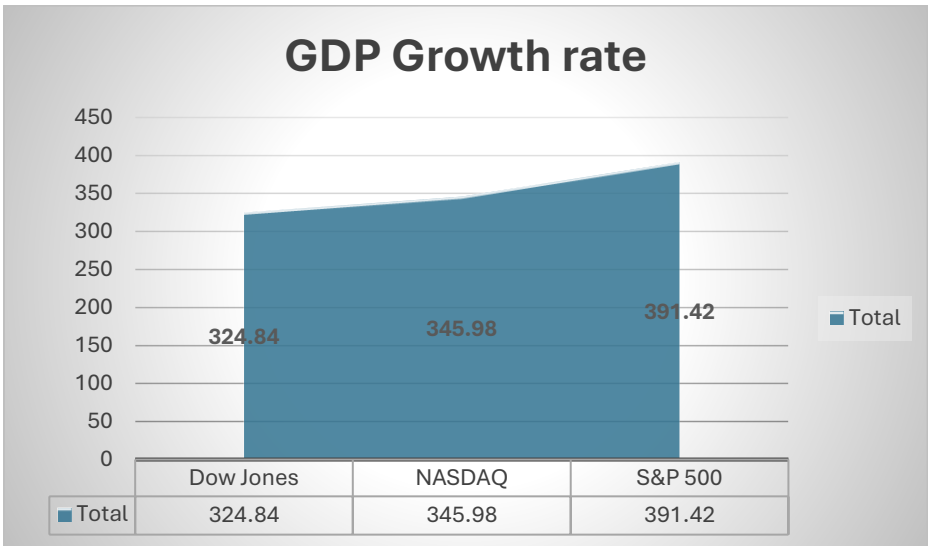
➤ Consumer & Retail KPIs

- Retail Sales Growth
- Consumer Spending Growth
- Spending vs Inflation Comparison

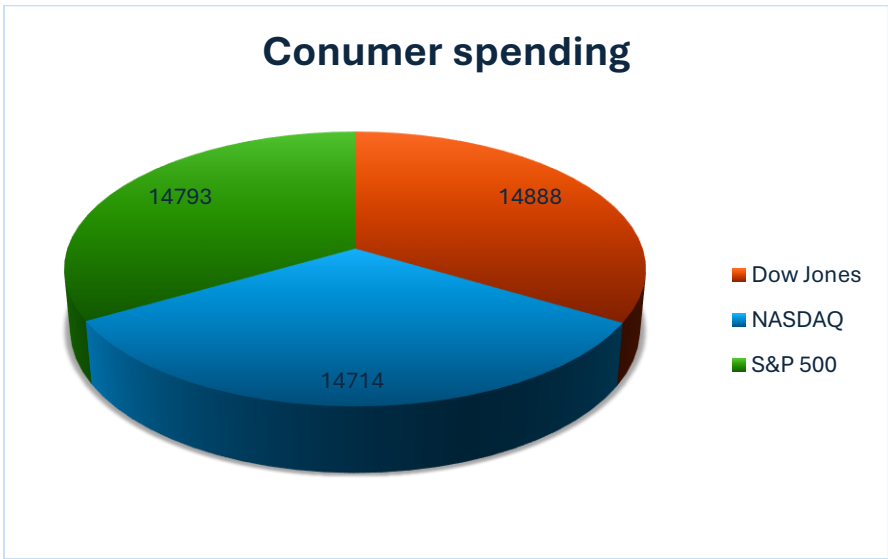


Data Visualisations

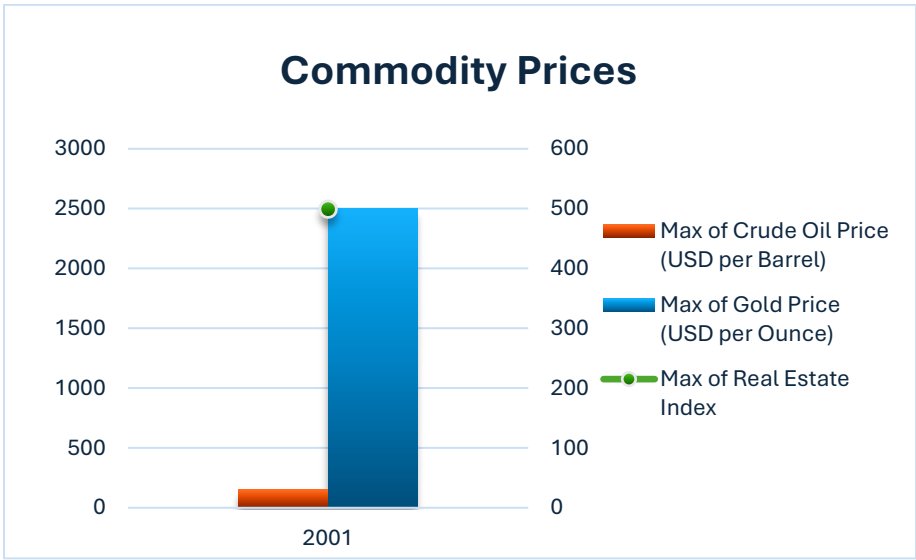
1.GDP Growth rate



2.Consumer Spending



3.Commodity Prices





Business Insights

- ✚ GDP growth increases market performance and profits.
- ✚ High inflation reduces consumer purchasing power.
- ✚ Rising oil prices increase business costs.
- ✚ Higher consumer spending supports economic growth.
- ✚ Interest rate changes affect borrowing and investment.
- ✚ Gold prices rise during economic uncertainty.

From the analysis, I observed that higher GDP growth leads to increased consumer spending and improved stock market performance.